

Technical Specification: Volumetric Weight Shipping Logic

This document specifies how to implement Kingz Logistics shipping calculations, ensuring the system bills for Volumetric Weight (VW) rather than physical weight when VW is higher.

1. The Logic Requirements

Kingz Logistics determines "Billable Weight" by comparing the physical scale weight (lbs) and the calculated volumetric weight (VW). The system must select the higher of the two values to calculate the final shipping fee.

2. Implementation Steps

1. Calculate Physical Weight (lbs).
2. Calculate Volumetric Weight (VW) using the formula: $(\text{Length} \times \text{Width} \times \text{Height}) / \text{Dimensional Factor}$.
3. Determine Billable Weight: $\text{Max}(\text{Physical Weight}, \text{Volumetric Weight})$.
4. Calculate Fee: $\text{Billable Weight} \times \$5.50/\text{lb}$.

3. Developer Code Implementation

```
function calculateKingzShipping(length, width, height, physicalWeightLb) {
    // Kingz Logistics rate per pound
    const ratePerLb = 5.50;

    // Dimensional factor (standard divisor)
    const dimFactor = 139;

    // Calculate Volumetric Weight (VW)
    let volumetricWeight = (length * width * height) / dimFactor;

    // Kingz Logic: Compare VW and physical LBS, take the higher number
```

```
    let billableWeight = Math.max(physicalWeightLbs, volumetricWeight);

    // Calculate final shipping fee
    let shippingFee = billableWeight * ratePerLb;

    return shippingFee.toFixed(2);
}
```